THEORY OF GAMES

INTRODUCTION

Generally, game refers to a situation of competition and conflict in which two or more competitors (or participants) are involved in decision-making process in anticipation of certain outcomes over a period of time.

NOTE: Competitors (participants) can also be described as players. Also, a player could be an individual, individuals or organizations.

Before it came into existence in the 20th century, John Von Neumann and Oscar Morgeristern in 1944 published book named Theory of Games and Economic Behaviour in which they emphasized on how business units of all categories can use this technique in determining the best strategies among various alternatives when found in a competitive business environment. Their approach was based on the principles of "best out of the worst"

For academic's purpose; theory of games provides series of mathematical models that may be useful in explaining interactive decision-making concepts where two or more competitors are involved under conditions of conflict and competition.

TERMS

1. **Number of Players**: If a game involves only two players (competitors) then it is called a two-person game. However, if the numbers of players are more than two, the game is referred to as n-person game.
2. **Sum of Gains and Losses**: If in a game, the sum of gains to one player (A) is exactly equal to the sum of losses to another player (B), so that the sum of gains (+12) and losses (-12) equals zero (0); then the game is said to be a zero-sum game. Otherwise, it is said to be non-zero-sum game.
3. **Two-Person Zero-Sum Games**: This is a game with only two players; say A and B and it is called a two-person zero sum game only if one player's gain is equal to the losses of the other player, so that total sum is zero.
4. **Payoff Matrix**: Pay-off is a quantitative measure of satisfaction that a player gets at the end of the play in terms of gains or losses, when players select their particular strategies

(course of action). Thus, payoff matrix is a form of matrix that depicts the payoffs of players involved in a play (game).

PAYOFF MATRIX

Player B's strategies

Player A's strategies	B_1	B_2	...	B_n
A_1	a_{11}	a_{12}	...	a_{1n}
A_2	a_{21}	a_{22}	...	a_{2n}
.	.	.		.
.	.	.		.
.	.	.		.
A_m	a_{m1}	a_{m2}	....	a_{mn}

Where: A_1, A_2, ..., A_m are player A's strategies

$B_1, B_2, ..., B_n$ are player B's strategies

$a_{11}, a_{12}, ..., a_{mn}$ are pay off

STRATEGY

These are course of actions that a player adopts for every payoff (outcome). The strategy for a player is the list of all possible actions (moves, decision alternatives courses of actions) that are likely to be adopted by him for every pay off (outcome).

TYPES

1. **Pure Strategy**: This is the only course of action that is always chosen by a player regardless of other player's strategy. The objective of the player is to maximise his/her gains or minimize the losses.
2. **Mixed Strategy**: This is a set of strategies that a player chooses on a particular move of the game with some fixed probability. Thus, there exist a probabilistic situation and players are opportune to maximize expected gain or minimize expected loss as their objective by making the choice among pure strategies with fixed probabilities.

ASSUMPTIONS OF THE GAME

I. Each player has a finite number of possible strategies available to him.

II. Players act rationally and intelligently.

III. List of strategies of each player and the amount of gain or loss on an individual's choice of strategy is known to each player in advance.

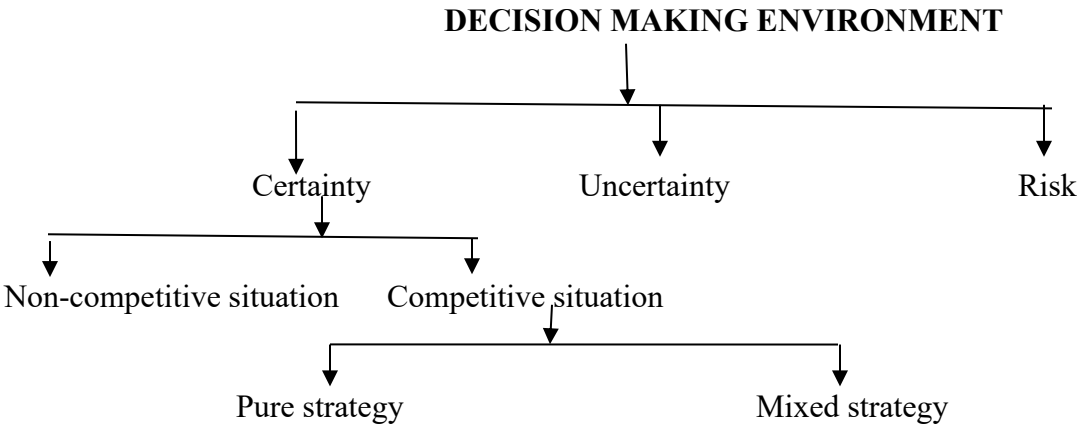
IV. One player attempts to maximize gains and the other attempts to minimize losses.

V. Players make decisions individually without any direct communication.

VI. Both players select and announce their strategies simultaneously so that neither player has an advantage resulting from direct knowledge of the other player's decision.

VII. Payoff is fixed and determined in advance.

NOTE:



SOLUTION OF GAME PROBLEM

PURE STRATEGIES

With the selection of an optimal strategy being a problem for each player thus, our objective is to know how these players must select their respective strategies so that they are able to optimize their payoff.

Pure Strategy Principles

- Maximin principle: Here, a player (A) selects a strategy that gives the largest gain among the row minimum values. The choice of player A is called the maximum principle and the corresponding gain is called the maximin value of the game.

- Minimax principle: Here, players (B) select the strategy that gives the minimum loss among the column maximum values. This choice of player B is called the minimax principle and the corresponding loss is the minimax value of the game.
- Value of the Game: This is the expected gain or loss in a game when a game is played large number of times. It is the amount of payoff 'V' at an equilibrium point.

NOTE:

I. The value of the game in general satisfies the equation:

$$\text{maximin value} \leq V \leq \text{minimax value}.$$

II. A game is said to be 'FAIR' game, if the lower (maximin) and upper (minimax) value of the game are equal and both equals zero.

III. A game is said to be strictly determinable if the lower (maximin) and upper (minimax) values of the game are equal and both equal the value of the game.

- Optimal Strategy: A course of actions that puts any player in the most preferred position irrespective of the course of action his competitor(s) adopt. Also, if the maximin value = the minimax value then the game is said to have a saddle (equilibrium) point and the corresponding strategies are called optimal strategies.
- Saddle Point: This is the payoff value that represent both minimax and maximin value of the game.

Example 1:

A company management and NLC are negotiating a new three-year settlement. Each of these has four strategies:

1. Hard and aggressive bargaining
2. Legalistic strategy
3. Reasoning and logical approach
4. Conciliatory approach

The costs to the company are given for every pair of strategy choice.

Company Strategies

Union strategies	I	II	III	IV
I	20	15	12	35
II	25	14	8	10
III	40	2	10	5
IV	-5	4	11	0

FIND: What strategy will the two sides adopt? Also determine the value of the game.

Solution:

Minimax-maximin principle

- From the union strategies, find the less pay off value on each row and write it under the column with “row minimum” respectively. Then, select the row value with the maximum value out of the less payoff identified.
- The identified value is the maximin value for the union while, its corresponding strategy would be the union’s chosen strategy for such negotiation.
- From the company strategies, find the highest payoff value on each column and write it under each column as column maximum. Then, select the column value with the minimum value out of the maximum value identified.
- The identified value is the minimax value for the company while, its corresponding strategy would be the company’s strategy for such negotiation.
- If the maximin and the minimax value are the same and meets at an equilibrium point, the value at the intersection point is the value of the game.

Company strategies

Union strategies	I	II	III	IV	Row minimum
I	20	15	12	35	12 maximin
II	25	14	8	10	8
III	40	2	10	5	2
IV	-5	4	111	0	0
Column	40	15	12 minimax	35	

maximum					

Strategy: Union Strategy = Strategy I

Company Strategy = Strategy III

Value = 12, since maximin value = minimax value

Example 2: Consider the game with the following payoff table:

		Player B	
Player A		B1	B2
	A1	2	6
	A2	-2	λ

a. Show that the game is strictly determinable, whatever may be.

b. Determine the value of the game.

Solution:

		Player B		
Player A		B1	B2	
	A1	2	6	[2] -- maximin
	A2	-2	λ	-2
		(2)	6	
		Minimax		

Value = 2 = saddle point

Since the saddle point is equal Value, the game is strictly determinable.

Example 3:

		Player B				
Player A		B1	B2	B3	B4	B5
	A1	-2	0	0	5	3
	A2	3	2	1	2	2
	A3	-4	-3	0	-2	6
	A4	5	3	-4	2	6

Find: Optimal strategy, Value, and Saddle Point.

Solution:

		Player B					
Player A		B1	B2	B3	B4	B5	Row minimum
	A1	-2	0	0	5	3	-2
	A2	3	2	1	2	2	1 ----- maximin value
	A3	-4	-3	0	-2	6	-4
	A4	5	3	-4	2	6	-4
Column maximum		5	3	1	5	6	
				Minimax			

Optimal strategy: PA – A2

PB – B3

Value of the game: 1

Example 4:

		Player B			
Player A		B1	B2	B3	B4
	A1	3	-5	0	6
	A2	-4	-2	1	2
	A3	5	4	2	3

Solution:

	Player B				
Player A	B1	B2	B3	B4	Row minimum
A1	3	-5	0	6	-5
A2	-4	-2	1	2	-4
A3	5	4	2	3	2 ------ Maximin
Column maximum	5	4	2	6	
			Minmax		

Optimal strategy: PA –A3

PB –B3

Value of the game = 2

MIXED STRATEGIES: GAME WITHOUT SADDLE POINT

This strategy is put into use when there is no equilibrium point i.e maximin value ≠ minimax value. Hence, the value of game obtained by the use of mixed strategies represent the least payoff which player A can expect to win and the least which player B can expect to lose.

The expected pay off to a player in a game with payoff matrix (aij)of order 'm*n' is defined as:

E (p,q) = $\sum_{i=1}^{m} \sum_{j=1}^{n}$ pi aij qj $= P^T$AQ

Where P= (P1, P2,''' Pm) and Q = (q1,q2, '''qn)

P1+P2+'''Pm=1 Q1+ Q2+ ''' Qn=1

MIXED STRATEGIES

I. ALGEBRAIC METHOD
II. GRAPHICAL METHOD
III. MATRIX METHOD
IV. ANALYTICAL METHOD
V. LINEAR PROGRAMMING

NOTE: Solving a 2*2 game without saddle point. That is,

	Player B	
Player A	a11	a12
	a21	a22

This formula can be used to find the value of game and optimal strategies.

$$P1 = \frac{a22 - a21}{a11 + a22 - (a12 + a21)}$$

$$Q1 = \frac{a22 - a21}{a11 + a22 - (a12 + a21)}$$

Where $P2 = 1 - P1$;

$Q2 = 1 - Q1$ and

$$V = \frac{a11\,a22 - a21a12}{a11 + a22 - (a12 + a21)}$$

Example 5:

In a game of matching coins with 2-players, suppose A wins 1 unit of value when there 2 heads, wins nothing when there are 2 tails, and loses 0.5 unit of value when there are 1 head and 1 tail. Determine:

- the payoff matrix
- The best strategy for each player, and
- The value of the game.

Solution

- Pay off matrix

		Player B	
Player A		H	T
	H	1	-0.5
	T	-0.5	0

- The optimal strategy for the players can be denoted by:

	A_1	A_2
$S_A =$	P_1	P_2

	B_1	B_2
$S_B =$	Q_1	Q_2

$$P_1 = \frac{a22 - a21}{a11 + a22 - (a12 + a21)} = \frac{0 - (-0.5)}{1 + 0 - (-0,5 + (-0,5))} = \frac{0.5}{1+1} = 1/4$$

$P_2 = 1 - ¼ = ¾$

$$Q_1 = \frac{a22 - a12}{a11 + a22 - (a12 + a21)} = \frac{0 - (-0,5)}{1 + 0 - (-0,5 + (-0,5))} = ¼$$

$Q_2 = 1 - ¼ = ¾$

The optimal strategy mixture

	A_1	A_2
$S_A =$	¼	¾

	B_1	B_2
$S_B =$	1/4	3/4

- The value of the game

$$V = \frac{a11a22 - a21a12}{a11 + a22 - (a12 + a21)} = \frac{1(0) - (-0.5(-0.5)}{1 + 0 - (-0,5 + (-0,5))} = \frac{-1/4}{2} = \frac{-1}{8}$$

Exercise:

Solve the game whose pay off matrix is given by

		Player B		
Player	A	B1	B2	B3
	A1	-2	-15	-2
	A2	-5	-6	-4
	A3	-5	20	-8

Determine:

The best strategy for each player, and

The value of the game.

Solve the game with the following matrix:

- $\begin{matrix} 5 & 1 \\ 3 & 4 \end{matrix}$

- $\begin{matrix} 4 & -4 \\ -4 & 4 \end{matrix}$

Determine:

The best strategy for each player, and

The value of the game.